

STABLE RANK ONE FOR COUNTABLE $C(K)$ CALKIN LIFTS

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ABSTRACT. We answer the countable compact metric $C(K)$ -Calkin question raised in Remark 2.23 of Horvath's paper arXiv:1807.10578. If X is a complex Banach space and K is countable compact metric with $\mathcal{B}(X)/\mathcal{K}(X) \cong C(K)$, then $\mathcal{B}(X)$ has stable rank one. Hence the Motakis–Puglisi–Zisimopoulou spaces X_K cannot provide examples for which $\mathcal{B}(X_K)$ is Dedekind-finite but fails stable rank one.

1. SOURCE QUESTION

Horvath studies the relation between stable rank one and Dedekind-finiteness for algebras of operators on Banach spaces. After proving that Tarbard's space gives a complex example with $\mathcal{B}(X)$ Dedekind-finite but not stable-rank-one, the paper considers the Motakis–Puglisi–Zisimopoulou construction. For every countable compact metric space K , their construction gives a Banach space X_K with

$$\mathcal{B}(X_K)/\mathcal{K}(X_K) \cong C(K).$$

Remark 2.23 observes that $\mathcal{B}(X_K)$ has left stable rank 1 or 2, and then asks whether some such K may still make $\mathcal{B}(X_K)$ Dedekind-finite while not having stable rank one.

2. PROOF INTUITION

The source already supplies one side of the extension: for every Banach space X , the unitization of $\mathcal{K}(X)$ has stable rank one. So if $\mathcal{B}(X)/\mathcal{K}(X) \cong C(K)$, the ideal and the quotient both have stable rank one as soon as $C(K)$ does.

The only missing point is whether this stable-rank-one information passes from ideal and quotient to the whole algebra. It does when quotient invertibles lift. For zero-dimensional compact K , $C(K)$ has path-connected invertible group: a nonvanishing continuous function has a continuous logarithm obtained by refining a finite cover to a clopen partition. Connected invertibles in the quotient lift by exponentials. A standard perturbation argument inside $1 + \mathcal{K}(X)$ then gives density of invertibles in $\mathcal{B}(X)$.

Remark 2.23. Let K be a countable compact metric space. By a deep result [12, Theorem B] of Motakis, Puglisi and Zisimopoulou, there exists a Banach space X_K such that $\mathcal{B}(X_K)/\mathcal{K}(X_K) \simeq C(K)$. In [16], Rieffel introduced the notion of *left stable rank* in general for unital Banach algebras. We observe here that $\mathcal{B}(X_K)$ has left stable rank 1 or 2, in the sense of Rieffel. By [16, Corollary 4.12], to show this, it is enough to observe that $\tilde{\mathcal{K}}(X_K)$ and $C(K)$ have both stable rank one. The former follows from Proposition 2.7. For the latter, in view of [16, Proposition 1.7], it is enough to see that K is zero-dimensional. This in particular follows if K is totally disconnected, by [1, Proposition 3.1.7]. But this is clearly holds, since K is countable and compact. We do not know however if for certain K -s the Banach algebra $\mathcal{B}(X_K)$ is DF but does not have stable rank one; our “lifting invertible elements” method is not applicable here, since K is totally disconnected.

After reading the first draft of this paper, it was suggested to us by Piotr Koszmider that the $C(K)$ -space from [15] is an example of a *real* Banach space such that its algebra of operators is Dedekind-finite but it does not have stable rank one. With his kind permission we include a proof here.

In the following, if K is a compact Hausdorff space, we always consider $C(K)$ as a *real* Banach space. To emphasise this, we write $C(K, \mathbb{R})$ for $C(K)$. Let K be a compact Hausdorff space, let $g \in C(K, \mathbb{R})$ then

$$M_g : C(K, \mathbb{R}) \rightarrow C(K, \mathbb{R}); \quad f \mapsto fg \quad (2.7)$$

is called the *multiplication operator*. An operator $T \in \mathcal{B}(C(K, \mathbb{R}))$ is called a *weak multiplication*

FIGURE 1. Source crop from Horvath, Remark 2.23, containing the countable compact metric $C(K)$ -Calkin question.

3. STABLE-RANK EXTENSION LEMMAS

We use stable rank one in Rieffel’s sense: a unital Banach algebra A has $\text{tsr}(A) = 1$ if $\text{inv}(A)$ is norm-dense in A .

Lemma 1 (Connected quotient invertibles lift). *Let A be a unital Banach algebra, $J \triangleleft A$ a closed two-sided ideal, and $q : A \rightarrow A/J$ the quotient map. If $\text{inv}(A/J)$ is connected, then every invertible element of A/J has an invertible lift in A .*

Proof. In a unital Banach algebra, the connected component of the identity in the invertible group consists of finite products of exponentials. Since $\text{inv}(A/J)$ is connected, each $b \in \text{inv}(A/J)$ has the form

$$b = \exp(b_1) \cdots \exp(b_n)$$

for some $b_i \in A/J$. Choose $a_i \in A$ with $q(a_i) = b_i$. Then

$$u = \exp(a_1) \cdots \exp(a_n)$$

is invertible in A , and $q(u) = b$. □

Lemma 2 (Stable-rank-one extension). *Let A be a unital Banach algebra and let $J \triangleleft A$ be a closed two-sided ideal. Suppose:*

- (1) *the unitization $\tilde{J}$ has stable rank one;*
- (2) *A/J has stable rank one;*
- (3) *every invertible element of A/J lifts to an invertible element of A .*

Then A has stable rank one.

Proof. Let $a \in A$ and let $\varepsilon > 0$. Since A/J has stable rank one, choose $b \in \text{inv}(A/J)$ with

$$\|q(a) - b\| < \delta,$$

where $\delta > 0$ will be chosen below. By the lifting hypothesis, choose $u \in \text{inv}(A)$ with $q(u) = b$. The quotient norm estimate gives some $j \in J$ such that

$$\|a - u - j\| < \delta.$$

Now $1 + u^{-1}j \in \tilde{J}$. Since $\tilde{J}$ has stable rank one, choose $v \in \text{inv}(\tilde{J})$ with

$$\|v - (1 + u^{-1}j)\| < \frac{\varepsilon}{2\|u\|}.$$

The element v , viewed in A , is invertible in A , hence uv is invertible in A . Moreover

$$\|uv - (u + j)\| \leq \|u\| \|v - (1 + u^{-1}j)\| < \varepsilon/2.$$

Choosing $\delta < \varepsilon/2$ gives

$$\|a - uv\| \leq \|a - u - j\| + \|u + j - uv\| < \varepsilon.$$

Thus invertibles are dense in A . □

Lemma 3 (Zero-dimensional $C(K)$). *Let K be a compact Hausdorff zero-dimensional space. Then $C(K)$ has stable rank one and $\text{inv}(C(K))$ is path-connected.*

Proof. We use the standard fact that every finite open cover of a compact zero-dimensional Hausdorff space has a finite pairwise disjoint clopen refinement.

First let $f \in C(K)$ and $\varepsilon > 0$. For each $z \in f(K)$, choose a nonzero scalar c_z and a neighborhood U_z of z in $\mathbb{C}$ such that $|w - c_z| < \varepsilon$ for all $w \in U_z$. Choose finitely many $U_{z_1}, \dots, U_{z_m}$ covering the compact set $f(K)$, and refine the open cover $f^{-1}(U_{z_i})$ to a finite clopen partition $E_1, \dots, E_r$, each E_ℓ contained in some $f^{-1}(U_{z_i})$. Define $g = c_{z_i}$ on E_ℓ whenever $E_\ell \subseteq f^{-1}(U_{z_i})$. Then g is continuous, nonvanishing, and $\|f - g\| < \varepsilon$. Hence $\text{inv}(C(K))$ is dense.

Now let $f \in \text{inv}(C(K))$. Cover the compact set $f(K) \subseteq \mathbb{C} \setminus \{0\}$ by finitely many open disks on which a branch of the logarithm is defined. After refining the corresponding inverse-image cover by a finite clopen partition, define a continuous function $h \in C(K)$ by using the chosen logarithm branch on each clopen piece. Then $\exp(h) = f$, so

$$t \mapsto \exp(th) \quad (0 \leq t \leq 1)$$

is a path in $\text{inv}(C(K))$ from 1 to f . Thus $\text{inv}(C(K))$ is path-connected. □

4. MAIN RESULT

Theorem 1. *Let X be a complex Banach space and let K be a countable compact metric space. If*

$$\mathcal{B}(X)/\mathcal{K}(X) \cong C(K)$$

as unital Banach algebras, then $\mathcal{B}(X)$ has stable rank one.

Proof. A countable compact metric space is zero-dimensional. Indeed, it is scattered, hence totally disconnected, and compact metric totally disconnected spaces are zero-dimensional. Therefore Lemma 3 gives

$$\text{tsr}(C(K)) = 1 \quad \text{and} \quad \text{inv}(C(K)) \text{ is connected.}$$

Set $A = \mathcal{B}(X)$ and $J = \mathcal{K}(X)$. Horvath’s Proposition 2.7 states that if J is a closed two-sided ideal of $\mathcal{B}(X)$ contained in the inessential operators, then $\widetilde{J}$ has stable rank one. Since compact operators are inessential, $\widetilde{\mathcal{K}(X)}$ has stable rank one.

The quotient A/J is isomorphic to $C(K)$, so A/J has stable rank one and connected invertible group. By Lemma 1, every invertible in A/J lifts to an invertible in A . Lemma 2 now applies and yields $\text{tsr}(\mathcal{B}(X)) = 1$. $\square$

Corollary 1. *For the Motakis–Puglisi–Zisimopoulou spaces X_K with K countable compact metric, $\mathcal{B}(X_K)$ cannot be Dedekind-finite while failing stable rank one. In fact $\mathcal{B}(X_K)$ has stable rank one.*

Proof. The MPZ construction gives $\mathcal{B}(X_K)/\mathcal{K}(X_K) \cong C(K)$, so Theorem 1 applies. Stable rank one implies Dedekind-finiteness, as recalled in Horvath’s Lemma 2.1, so the non-stable-rank-one alternative in Remark 2.23 is impossible. $\square$

5. VERIFICATION AND NOVELTY CHECK

The proof is non-computational. Its dependencies are:

- Horvath’s Proposition 2.7, giving stable rank one for $\widetilde{\mathcal{K}(X)}$;
- the standard zero-dimensional topology facts used in Lemma 3;
- the elementary Banach-algebra fact that the identity component of the invertible group consists of finite products of exponentials.

The cheap run indexes were searched for ‘1807.10578’, ‘Dedekind-finite’, ‘stable rank’, ‘ X_K ’, ‘MPZ’, ‘Calkin’, and countable compact $C(K)$ phrases. No exact existing packet was found. Local source checks included Horvath’s paper and Motakis’s later arXiv:2110.10868 paper, whose complemented-subspace section gives related Dedekind-finiteness information but not this stable-rank extension observation. A bounded web search for exact phrase combinations around $\mathcal{B}(X_K)$, stable rank one, $C(K)$, Calkin algebras, and arXiv:1807.10578 found no exact prior answer. Because the argument is short, the novelty confidence is moderate rather than absolute.

REFERENCES

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